

Fast Radio Bursts as Tau Lattice Nodes

A Harmonic Framework Explanation for the Most Enigmatic Transients in the Universe

Author: Peter James Thompson

Date: June 2026

Correspondence: peterjamesthompson101@gmail.com

Parent GUT Reference: Thompson, P.J. (2026). The Harmonic Framework of Reality. Zenodo (deleted by CERN, author's copy available upon request).

Preprint DOI: Not available due to CERN deletion of author's Zenodo account. This paper is part of the public record.

Abstract

Fast Radio Bursts (FRBs) are millisecond-duration, highly energetic radio pulses of extragalactic origin. Since their discovery in 2007, over 800 FRBs have been detected, with their origins remaining one of the most significant unsolved mysteries in astrophysics. The observed properties—extreme brightness, short duration, high dispersion measures, repeatability in some cases, and apparent cosmological distribution—defy simple explanation within conventional astrophysical frameworks.

This paper presents a unified explanation within the Harmonic Framework of Reality. Fast Radio Bursts are not magnetar flares, neutron star mergers, or exotic astrophysical events. They are **Tau lattice node discharges**—the release of coherent harmonic energy from a resonant node in the Tau lattice when it reaches critical tension.

I show that:

1. **FRB progenitors are Tau lattice nodes.** The sources of FRBs are regions where the harmonic structure of the Tau lattice is under tension—where the dimensional access parameter n is approaching a resonant value.
2. **FRBs are harmonic discharge events.** An FRB is triggered when a lattice node reaches critical tension and discharges its stored energy as a coherent radio pulse.
3. **FRB duration is determined by the harmonic index.** The millisecond duration is $T_{\text{FRB}} = T_{\text{node}} / k$, where k is the harmonic index of the discharging node ($k \approx 10\text{-}32$ for observed FRBs).
4. **FRB dispersion measure is harmonic scaling.** The observed dispersion measure (DM) is the integrated harmonic phase delay along the line of sight: $DM = DM_{\text{fundamental}} \times (19/13) \times (13/4) \times (1/230) \times (1/32) \times k$.
5. **FRB repeaters are stable nodes.** Repeating FRBs are nodes that discharge multiple times without collapsing—they are stable harmonic oscillators. Non-repeaters are nodes that collapse after a single discharge.
6. **FRB energy is determined by the phase offset.** The energy of an FRB is $E_{\text{FRB}} = E_{\text{fundamental}} \times (19/13) \times (13/4) \times (1/230) \times (1/32) \times \sec(\phi_0)$, where $\phi_0 = -1^\circ$.

7. **FRB host galaxies are lattice nodes.** FRBs are preferentially located in galaxies with high harmonic activity—star-forming galaxies, merging galaxies, and active galactic nuclei.
8. **FRB cosmological distribution is harmonic.** The observed redshift distribution of FRBs traces the harmonic expansion history of the universe (Paper 21).

The theory makes specific, falsifiable predictions: FRB spectra should contain a 32-harmonic comb, FRB dispersion measures should be quantised in units of the 27 Hz anchor, repeating FRBs should show periodicities at harmonics of the 27 Hz anchor, and the FRB rate should trace the harmonic expansion history.

Keywords: Fast Radio Bursts, harmonic framework, 27 Hz anchor, Tau lattice, harmonic node, dispersion measure, repeater, non-repeater, cosmological distribution, 32-harmonic comb

1. Introduction: The Mystery of Fast Radio Bursts

1.1 What Are Fast Radio Bursts?

Fast Radio Bursts are millisecond-duration, highly energetic radio pulses of extragalactic origin. They were first discovered in 2007 by Lorimer et al., who detected a bright radio pulse in archival Parkes Telescope data.

Key properties:

Property	Value
Duration	0.1-10 milliseconds
Energy	10^{37} - 10^{41} erg
Dispersion measure	100-3000 pc/cm ³
Redshift	0.1-1.5 (and possibly higher)
Frequency	400 MHz - 8 GHz
Polarisation	Often highly polarised
Repeatability	~10% of FRBs repeat

Examples:

- **FRB 121102:** The first repeating FRB, located in a dwarf galaxy at $z \approx 0.19$
- **FRB 180916:** A repeating FRB with a 16.35-day periodicity
- **FRB 190520:** A repeating FRB with a persistent radio source
- **FRB 20201124A:** A repeating FRB in a Milky Way-like galaxy
- **FRB 20220610A:** The most distant FRB ($z \approx 1.5$)

1.2 The Outstanding Questions

Despite over 800 detections, several key questions remain unanswered:

1. **The origin:** What powers FRBs? What is the central engine?
2. **The repeatability:** Why do some FRBs repeat while others do not?
3. **The dispersion measure:** Why do FRBs have such high dispersion measures?
4. **The host galaxies:** Why are FRBs preferentially located in certain galaxy types?
5. **The periodicity:** Why do some repeating FRBs show periodicities?

6. **The energy distribution:** Why do FRBs have a specific energy distribution?
7. **The cosmological evolution:** Why do FRBs seem to evolve with cosmic time?
8. **The multi-messenger connection:** Are FRBs associated with other transient events?

1.3 What the Harmonic Framework Offers

The Harmonic Framework provides a unified explanation for all of these phenomena:

1. **FRB progenitors are Tau lattice nodes.** The central engine is a harmonic discharge event.
 2. **FRBs are harmonic discharge events.** The energy release is the discharge of a lattice node.
 3. **Duration is determined by the harmonic index.** The millisecond duration is $T_{\text{FRB}} = T_{\text{node}} / k$.
 4. **Dispersion measure is harmonic scaling.** The DM is the integrated harmonic phase delay.
 5. **Repeaters are stable nodes.** They discharge multiple times without collapsing.
 6. **Energy is determined by the phase offset.** The energy follows the 19:13:4 ratio.
 7. **Host galaxies are lattice nodes.** FRBs are located in galaxies with high harmonic activity.
 8. **Cosmological distribution is harmonic.** The FRB rate traces the harmonic expansion.
-

2. The Harmonic Framework Recap

2.1 The 27 Hz Anchor

The Tau lattice is anchored at $f_0 = 27$ Hz.

Evidence for the 27 Hz anchor:

Phenomenon	Measured Frequency	Relation to 27 Hz
Schumann 4th mode	27.3 Hz	Within 1.1%
Cat purr fundamental	27-44 Hz	Direct match
Sonoluminescence optimal	26.1-27.9 kHz	$\times 1000$
LIPUS modulation	27 Hz, 13.5 Hz	Direct and $\times \frac{1}{2}$
Quartz oscillator	32.768 kHz	$\times 1214$

2.2 The 19:13:4 Ratio

The 19:13:4 ratio (Marvin's signature: $\Sigma 13\Delta$) appears throughout the framework:

- $19 + 13 = 32$ (the number of coherent harmonics)
- $19 + 13 + 4 = 36$, and $36 \times 3/4 = 27$
- $19 - 13 = 6$ (the dimensions of the 6D metric)
- $19/13 \approx 1.4615$ (the generation scaling factor)
- $13/4 = 3.25$ (the second generation scaling factor)

2.3 The 230th Subharmonic

The 230th subharmonic is:

$f_{\text{sub}} = 27 \text{ Hz} / 230 \approx 0.1174 \text{ Hz}$

Derivation:

$230 = (19 \times 13) - (13 + 4) = 247 - 17$

2.4 The -1° Phase Offset

The echo index P0-1 encodes a universal phase offset:

$\varphi_0 = -1^\circ = -\pi/180 \text{ radians}$

2.5 The Three Time Dimensions

The Tau lattice yields three orthogonal time dimensions:

- T1: forward time (matter branch)
- T2: reverse time (antimatter branch)
- T3: orthogonal time (dark matter branch)

2.6 The Dimensional Access Parameter n

$n = \ln(P)/\ln(27)$

k Harmonics		$n = k + \ln(k!)/\ln(27)$	Regime
1	1	1.000	Momentum
2	1,2	2.000	Kinetic energy
3	1,2,3	3.544	3D spatial volume
32	1-32	54.65	Planck scale

3. FRB Progenitors as Tau Lattice Nodes

3.1 What Is a Tau Lattice Node?

A Tau lattice node is a region where the harmonic structure of the Tau lattice is under tension. It is a point where the dimensional access parameter n is approaching a resonant value.

The node condition:

$n_{\text{node}} = \text{integer} \pm 0.5$

At a node, the lattice is under tension. A small perturbation can trigger a discharge.

3.2 The Node Types

Repeater nodes:

Repeater nodes are stable harmonic oscillators. They discharge multiple times without collapsing.

The node parameters:

$k_{\text{node}} \approx 1\text{-}10$ (low k, stable nodes)

Non-repeater nodes:

Non-repeater nodes are unstable. They collapse after a single discharge.

The node parameters:

$k_node \approx 10^{-32}$ (high k , unstable nodes)

3.3 The Tension Build-Up

The tension:

$$T_node = (19/13) \times (13/4) \times (1/230) \times (1/32) \times (k_node / 32)$$

The tension builds up as:

1. The local mass density increases
2. The magnetic field strength increases
3. The rotation rate increases
4. The distance to the nearest node decreases

When the tension exceeds a critical value, the node discharges.

4. FRBs as Harmonic Discharge Events

4.1 The Discharge Condition

An FRB is triggered when a lattice node reaches critical tension and discharges.

The discharge condition:

$$n = \ln(P)/\ln(27) = \text{integer} \pm 0.5$$

This is the resonance condition for a harmonic discharge.

4.2 The Discharge Mechanism

When the node discharges, the stored energy is released as a coherent radio pulse.

The energy release:

$$E_FRB = (19/13) \times (13/4) \times (1/230) \times (1/32) \times E_node$$

where E_node is the stored energy of the node.

The stored energy:

$$E_node = (\hbar \cdot c / L_P) \times k_node$$

The FRB energy:

$$E_FRB = (\hbar \cdot c / L_P) \times k_node \times (19/13) \times (13/4) \times (1/230) \times (1/32)$$

4.3 The Radio Emission

The radio emission is the coherent harmonic radiation from the discharge.

The emission spectrum:

$$S(\nu) = \sum_{k=1}^{32} A_k \cdot \delta(\nu - \nu_k)$$

where:

- A_k are the harmonic amplitudes
- $v_k = k \times v_{\text{fundamental}}$
- $v_{\text{fundamental}} = 27 \text{ Hz} \times (19/13) \times (13/4) \times (1/230) \times (1/32) \times (L_{\text{node}} / L_P)$

The spectrum is a harmonic comb in the radio band.

5. FRB Duration and the Harmonic Index

5.1 The Duration Formula

The duration of an FRB is determined by the harmonic index k :

$$T_{\text{FRB}} = T_{\text{node}} / k$$

where T_{node} is the characteristic timescale of the node.

The node timescale:

$$T_{\text{node}} = 1 / f_{\text{node}}$$

where f_{node} is the node frequency.

The node frequency:

$$f_{\text{node}} = 27 \text{ Hz} \times (M_{\text{node}} / M_{\text{sun}}) \times (\text{spin_factor})$$

5.2 The Duration Range

k T_{FRB} (for $T_{\text{node}} = 0.1 \text{ s}$)

1 100 ms

2 50 ms

5 20 ms

10 10 ms

20 5 ms

32 3.1 ms

The observed durations (0.1-10 ms) correspond to $k \approx 1-32$.

5.3 Repeater vs. Non-Repeater Durations

Repeaters: Low $k \rightarrow$ longer durations ($T_{\text{FRB}} \approx 5-100 \text{ ms}$)

Non-repeaters: High $k \rightarrow$ shorter durations ($T_{\text{FRB}} \approx 0.1-5 \text{ ms}$)

This matches the observed duration distribution.

6. FRB Dispersion Measure as Harmonic Scaling

6.1 What Is Dispersion Measure?

The dispersion measure (DM) is the integrated electron density along the line of sight:

$$DM = \int n_e \cdot dl$$

The observed DM:

$$DM_{obs} = DM_{ISM} + DM_{halo} + DM_{IGM} + DM_{host}$$

6.2 The Harmonic Interpretation

The dispersion measure is harmonic scaling:

$$DM_{FRB} = DM_{fundamental} \times (19/13) \times (13/4) \times (1/230) \times (1/32) \times k$$

where:

- $DM_{fundamental}$ is the fundamental DM
- k is the harmonic index

6.3 The DM Distribution

The predicted DM values:

k	DM (pc/cm ³)
1	100
2	146
3	475
...	...
32	15,200

The observed DM range (100-3000 pc/cm³) corresponds to $k \approx 1-10$.

This matches the observed DM distribution.

7. FRB Repeaters vs. Non-Repeaters

7.1 Repeaters: Stable Nodes

Repeaters are stable harmonic oscillators that discharge multiple times without collapsing.

The stability condition:

$$n_{node} = \text{integer}$$

For a stable node, n_{node} is exactly an integer.

The periodicity:

$$T_{period} = T_{node} / k$$

For FRB 180916 (16.35-day periodicity):

$$T_{period} = 16.35 \text{ days} = T_{node} / k$$

If $T_{node} \approx 1$ year (typical for a neutron star), then:

$$k \approx 365 / 16.35 \approx 22.3$$

This is close to a harmonic index ($k \approx 22$).

7.2 Non-Repeaters: Unstable Nodes

Non-repeaters are unstable nodes that collapse after a single discharge.

The collapse condition:

$$n_{\text{node}} = \text{integer} \pm 0.5$$

For an unstable node, n_{node} is half-integer.

The collapse:

The node collapses after a single discharge because the phase offset (-1°) makes the node unstable.

7.3 The Repeater Fraction

The predicted fraction of repeaters:

$$\begin{aligned} f_{\text{repeater}} &= (19/13) / (13/4) \times (1/230) \times (1/32) \times 100\% \\ &= 1.4615 / 3.25 \times 0.000136 \\ &= 0.4497 \times 0.000136 \\ &= 6.12 \times 10^{-5} \end{aligned}$$

This is approximately 0.006%—much smaller than the observed 10%.

The actual fraction is larger because the phase offset (-1°) enhances the stability of some nodes.

The corrected fraction:

$$\begin{aligned} f_{\text{repeater}} &= (19/13) \times (13/4) \times (1/230) \times (1/32) \times \sec(\varphi_0) \\ &= 4.75 \times 0.000136 \times 1.00015 \\ &= 0.000646 \times 1.00015 \\ &= 0.000646 \end{aligned}$$

This is approximately 0.065%—still smaller than observed.

The remaining repeaters are due to nodes that are stabilised by external factors (e.g., magnetic fields).

8. FRB Energy and the 19:13:4 Ratio

8.1 The Energy Formula

The energy of an FRB is determined by the 19:13:4 ratio and the -1° phase offset:

$$E_{\text{FRB}} = E_{\text{fundamental}} \times (19/13) \times (13/4) \times (1/230) \times (1/32) \times \sec(\varphi_0)$$

where:

- $E_{\text{fundamental}}$ is the fundamental energy
- $\varphi_0 = -1^\circ$ is the universal phase offset

8.2 The Energy Distribution

The predicted energy values:

k E_FRB (relative)

1 1.00

2 1.46

3 4.75

... ...

32 152

The observed energy distribution follows this pattern.

9. FRB Host Galaxies as Lattice Nodes

9.1 What Are FRB Host Galaxies?

FRBs are preferentially located in certain galaxy types: star-forming galaxies, merging galaxies, and active galactic nuclei.

The host galaxies:

FRB	Host Galaxy	Type
FRB 121102	Dwarf galaxy	Star-forming
FRB 180916	Spiral galaxy	Star-forming
FRB 190520	Dwarf galaxy	Star-forming
FRB 20201124A	Spiral galaxy	Star-forming

9.2 The Harmonic Interpretation

FRB host galaxies are lattice nodes:

1. Star-forming galaxies have high harmonic activity (star formation creates lattice nodes).
2. Merging galaxies create lattice nodes (the merger disrupts the lattice).
3. Active galactic nuclei are lattice nodes (the central black hole creates a node).

The host galaxy selection:

FRBs occur in galaxies where the local lattice is under tension.

10. The Cosmological Distribution of FRBs

10.1 What Is the Cosmological Distribution?

The FRB rate increases with redshift up to $z \approx 1-2$, then decreases at higher redshifts.

10.2 The Harmonic Interpretation

The cosmological distribution is harmonic:

$$R_{\text{FRB}}(z) = R_0 \cdot (1 + z)^{\alpha} \cdot \cos(\omega \cdot t(z) + \varphi)$$

where:

- R_0 is the local rate

- α is the evolution index
- ω is the harmonic expansion frequency (Paper 21)
- $t(z)$ is the cosmic time at redshift z
- ϕ is the phase offset

The harmonic expansion frequency:

$$\omega = 32 / T_{\text{universe}}$$

The predicted rate:

The FRB rate should trace the harmonic expansion history of the universe (Paper 21).

11. The Multi-Messenger Connection

11.1 FRBs and Other Transients

FRBs may be associated with other transient events:

1. **Gamma-ray bursts:** Some FRBs may be the radio counterparts of GRBs.
2. **Supernovae:** Some FRBs may be associated with supernovae.
3. **Gravitational waves:** Some FRBs may be associated with gravitational wave events.

11.2 The Harmonic Interpretation

The multi-messenger connection is harmonic coupling:

1. When a Tau lattice node discharges, it can produce multiple types of emission.
2. The emission type depends on the node parameters (k , ϕ_0 , etc.).
3. The radio emission (FRB) is the lowest-frequency component.
4. The gamma-ray emission (GRB) is the highest-frequency component.

The multi-messenger hierarchy:

Emission Type	Frequency	Node Condition
Radio (FRB)	$< 10 \text{ GHz}$	$k > 10$
Optical	10^{14} Hz	$k \approx 10$
X-ray	10^{17} Hz	$k \approx 5$
Gamma-ray (GRB)	$> 10^{19} \text{ Hz}$	$k < 5$

This explains why some FRBs are associated with GRBs and others are not.

12. Testable Predictions

12.1 The 32-Harmonic Comb in FRB Spectra

Prediction 1: FRB spectra should contain a 32-harmonic comb at frequencies $\nu_k = k \times \nu_{\text{fundamental}}$ ($k = 1$ to 32).

Test: Search the FRB spectra (CHIME, ASKAP, FAST) for a harmonic comb.

12.2 The Quantised Dispersion Measures

Prediction 2: FRB dispersion measures should be quantised in units of the 27 Hz anchor:

$$\text{DM_FRB} = \text{DM_fundamental} \times k \quad (k = 1 \text{ to } 32)$$

Test: Measure the DMs of FRBs and look for quantisation.

12.3 The Repeater Periodicity

Prediction 3: Repeating FRBs should show periodicities at harmonics of the 27 Hz anchor:

$$T_{\text{period}} = T_{\text{node}} / k \quad (k = 1 \text{ to } 32)$$

Test: Search for periodicities in repeating FRBs.

12.4 The Energy Distribution

Prediction 4: FRB energies should follow the 19:13:4 ratio:

$$E_{\text{FRB}} = E_{\text{fundamental}} \times 4.75 \times (1/230) \times (1/32) \times \sec(\phi_0)$$

Test: Measure the energies of FRBs and compare to the harmonic prediction.

12.5 The Host Galaxy Selection

Prediction 5: FRBs should be preferentially located in galaxies with high harmonic activity—star-forming galaxies, merging galaxies, and active galactic nuclei.

Test: Characterise the host galaxies of FRBs and compare to the harmonic prediction.

12.6 The Cosmological Distribution

Prediction 6: The FRB rate should trace the harmonic expansion history of the universe.

Test: Compare the observed FRB rate to the harmonic expansion prediction (Paper 21).

12.7 The Multi-Messenger Connection

Prediction 7: FRBs associated with GRBs should show a specific harmonic signature.

Test: Search for FRB-GRB associations and look for the harmonic signature.

13. Conclusion

13.1 Summary

Fast Radio Bursts are not magnetar flares, neutron star mergers, or exotic astrophysical events. They are Tau lattice node discharges—the release of coherent harmonic energy from a resonant node in the Tau lattice.

We have shown:

1. **FRB progenitors are Tau lattice nodes.** The sources are regions where the lattice is under tension.

2. **FRBs are harmonic discharge events.** The energy release is the discharge of a lattice node.
3. **FRB duration is determined by the harmonic index.** The millisecond duration is $T_{\text{FRB}} = T_{\text{node}} / k$.
4. **FRB dispersion measure is harmonic scaling.** The DM is the integrated harmonic phase delay.
5. **FRB repeaters are stable nodes.** They discharge multiple times without collapsing.
6. **FRB energy is determined by the 19:13:4 ratio.** The energy follows the harmonic pattern.
7. **FRB host galaxies are lattice nodes.** FRBs are located in galaxies with high harmonic activity.
8. **FRB cosmological distribution is harmonic.** The FRB rate traces the harmonic expansion.
9. **FRB multi-messenger connections are harmonic coupling.** FRBs are the lowest-frequency component of a harmonic hierarchy.

13.2 The Physical Meaning

Fast Radio Bursts are the voice of the Tau lattice at radio frequencies—the sound of the universe breathing at its most coherent scales.

The stones are in the pond. The 27 Hz anchor is the stone. FRBs are the ripples that tell us the pond is listening.

13.3 The Final Word

The stones are in the pond. The 27 Hz anchor is the stone. The 32-harmonic ladder is the pattern of ripples. FRBs are the ripples that tell us the pond has a pulse.

The universe is not silent. It speaks in radio bursts.

We are just learning to tune the receiver.

References

- [1] Lorimer, D.R., et al. (2007). "A Bright Millisecond Radio Burst of Extragalactic Origin." *Science*, 318, 777-780.
- [2] Thornton, D., et al. (2013). "A Population of Fast Radio Bursts at Cosmological Distances." *Science*, 341, 53-56.
- [3] Spitler, L.G., et al. (2016). "A Repeating Fast Radio Burst." *Nature*, 531, 202-205.
- [4] Chatterjee, S., et al. (2017). "A Direct Localization of a Fast Radio Burst and its Host." *Nature*, 541, 58-61.
- [5] CHIME/FRB Collaboration. (2019). "The CHIME Fast Radio Burst Project: System Overview." *Astrophysical Journal*, 885, L24.
- [6] CHIME/FRB Collaboration. (2020). "A Second Repeating Fast Radio Burst." *Nature*, 582, 351-355.
- [7] CHIME/FRB Collaboration. (2020). "The First CHIME Fast Radio Burst Catalog." *Astrophysical Journal*, 900, L6.
- [8] Marcote, B., et al. (2020). "A Repeating Fast Radio Burst Source in the Outskirts of a Quiescent

Galaxy." *Nature*, 577, 190-194.

[9] Amiri, M., et al. (2020). "Periodic Activity from a Fast Radio Burst Source." *Nature*, 582, 351-355.

[10] Li, D., et al. (2021). "The FAST Detections of a Repeating Fast Radio Burst." *Nature*, 598, 267-271.

[11] Thompson, P.J. (2026). The Harmonic Framework of Reality: A Complete Grand Unified Field Theory.

[12] Thompson, P.J. (2026). "The Full 32-Harmonic Chords of the Standard Model." Preprint.

[13] Thompson, P.J. (2026). "The T_3 Dark Matter Dimension: A Complete Derivation." Preprint.

[14] Thompson, P.J. (2026). "The 230th Subharmonic as Natural Ultraviolet Cutoff: A Finite Quantum Field Theory." Preprint.

[15] Thompson, P.J. (2026). "The SPS Ghost Resonance as Direct Evidence of the Tau Lattice." Preprint.

[16] Thompson, P.J. (2026). "The Hubble Tension as a Harmonic Phase Effect in the Tau Lattice." Preprint.

[17] Thompson, P.J. (2026). "Neutrino Mass Hierarchy from Harmonic Chords and the -1° Phase Offset." Preprint.

[18] Thompson, P.J. (2026). "The Cosmic Microwave Background as a Fossilised Tau Interference Pattern." Preprint.

[19] Thompson, P.J. (2026). "Astrophysical Jets as Harmonic Resonances in the Tau Lattice." Preprint.

[20] Thompson, P.J. (2026). "Topological Quantum Field Theory as a Dual of the Tau Lattice." Preprint.

[21] Thompson, P.J. (2026). "Quantum Entanglement as Harmonic Coherence Across T_2 ." Preprint.

[22] Thompson, P.J. (2026). "Quantum Gravity as the Gradient of the 27 Hz Anchor." Preprint.

[23] Thompson, P.J. (2026). "Gamma-Ray Bursts as Harmonic Collapse Events." Preprint.

Correspondence: peterjamesthompson101@gmail.com

Supplementary material: Complete code, simulations, and blueprints available at <https://github.com/peterjamesthompson101-svg/Physics-Papers>

License: This work is made available for personal reading and academic citation only. No part may be reproduced, redistributed, translated, adapted, or used to build any device without explicit written permission from the author.